

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

PAPER-2

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

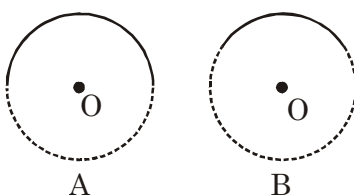
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

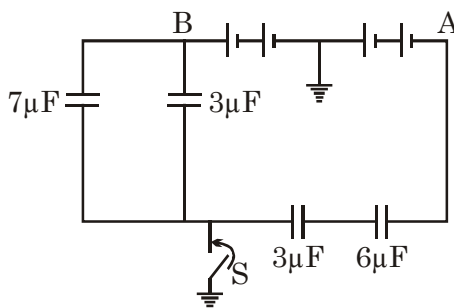
1. Structure-A is a half-circle infinite sheet carrying uniform electric current I flowing out of the paper plane. Structure-B is a quarter-circle infinite sheet carrying uniform electric current I flowing out of the paper plane.



- (A) At the center of the circular arc point-O the total magnetic field is pointing to right in both cases
- (B) At the center of the circular arc point-O the total magnetic field is pointing to left in both cases
- (C) The field strength of structure-A is larger than that of structure-B.
- (D) The field strength of structure-A is smaller than that of structure-B.

Space for Rough Work

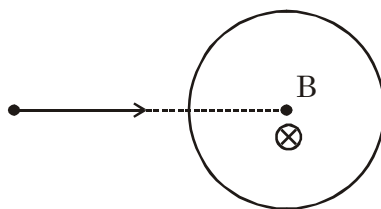
2. Magnetic field at origin is $2\hat{k}$ tesla. Which of the following current distribution could have caused it?
- (A) A current flowing on a plane parallel to xy plane. The current is flowing in +ve x direction.
Current / length = $\frac{4}{\pi} \times 10^7 \text{ A/m}$.
- (B) Current flowing in circular loop in xy plane centred at origin current is flowing anticlockwise as seen from +ve z direction. $i = 100 \text{ A}$, radius = $\pi \times 10^{-5} \text{ m}$.
- (C) Current of 10^7 A flowing in an infinite wire kept along line $x = +1$. Current flowing in +ve y direction
- (D) Current of $\frac{10^4}{2\pi} \text{ A}$ flowing in a solenoid having 1000 turns/m. The axis of solenoid is z-axis.
3. Potential of A is -60V and potential of B is $+60\text{V}$.



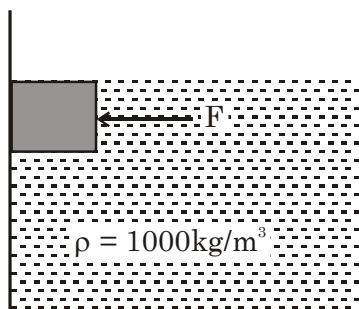
- (A) The charge on $6\mu\text{F}$ increases after switch is closed.
- (B) The charge on $7\mu\text{F}$ increases after switch is closed.
- (C) The charge on $6\mu\text{F}$ decreases after switch is closed.
- (D) The charge on $7\mu\text{F}$ decreases after switch is closed.

Space for Rough Work

4. Uniform magnetic field is confined into a cylindrical region, if α -particles with different speeds enter the region radially, then :



- (A) Faster the particle, lesser is the time spent in region.
 (B) Slower the particle, lesser is the time spent in region.
 (C) Slower the particle, greater is the time spent in region.
 (D) Time will be same for all particles
5. Which of the following statement is/are correct for mechanical standing wave on a stretched wire ?
 (A) Elastic potential energy of a small element at antinode is constant and minimum.
 (B) Elastic potential energy of a small element at node is constant and maximum.
 (C) Total energy of an element is constant.
 (D) Total kinetic energy between two consecutive nodes become maximum twice in one time period.
6. A cubical block of mass 5 kg and side 10 cm is pressed against a rough wall ($\mu = 0.9$) with a force F passing through the centre of cube inside a swimming pool as shown in the figure. Then :



- (A) The cube will remain in equilibrium if the force is $F \geq \frac{400}{9}$ Newton.
 (B) The cube will remain in equilibrium if the force is $F < \frac{355}{9}$ Newton.
 (C) The friction force acting on the cube is 40 N if $F = \frac{110}{3}$ N.
 (D) The friction force acting on the cube is 40 N if $F = 50$ N.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

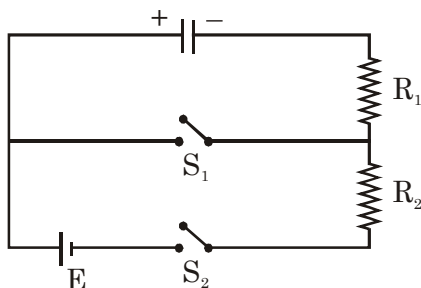
- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS. **For Example :** If answer is -77.25, 5.2 then fill the bubbles as follows.

										+	●
●	●	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

										●	-
●	●	●	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

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Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

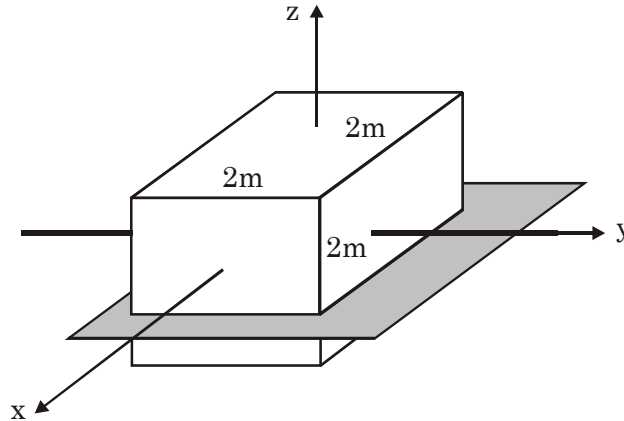
- The capacitor shown in figure has been charged to a potential difference of V volts, so that it carries a charge CV with both the switches S_1 and S_2 remaining open. Switch S_1 is closed at $t = 0$. At $t = R_1C$ switch S_1 is opened and S_2 is closed. Find the charge on the capacitor in μC at $t = 2R_1C + R_2C$. Take $V = eE$, $C = 4 \mu F$ and $E = 3 V$. (e here refers to natural base.)



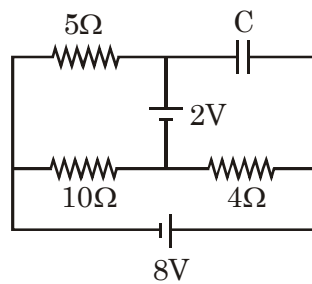
- Plate 1 of a capacitor having capacitance $1.5 \mu F$ is given a charge of $30 \mu C$ while the other plate is uncharged. The capacitor is now connected with a battery of potential difference $40 V$ with plate 1 connected to the positive terminal. Work done by battery is _____ μJ .

Space for Rough Work

3. Charge in the form of a plane sheet with density $\sigma = 40 \mu\text{C}/\text{m}^2$ is located at $z = -0.5 \text{ m}$. A uniform line of charge of density $\lambda = -6 \mu\text{C}/\text{m}$ lies along the y -axis. Net flux that crosses the surface of cube 2m on an edge, centered at the origin as shown, is _____ $\frac{10^{-6}}{\epsilon_0} \frac{\text{N}}{\text{C}} \text{m}^2$

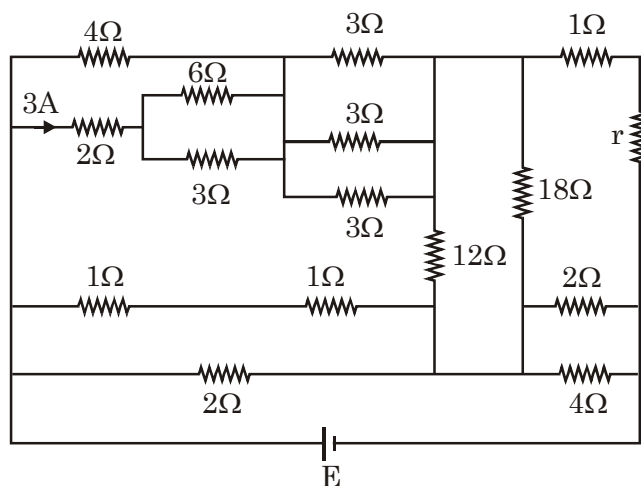


4. Calculate the potential difference across the capacitor in the circuit (in Volts).



Space for Rough Work

5. A 3m long organ pipe open at both ends is driven to third harmonic standing wave. The maximum amplitude of pressure oscillation is 0.1 % of the mean atmospheric pressure $P_0 = 10^5 \text{ N/m}^2$. The maximum amplitude of particle oscillation in metre is $\frac{\alpha}{1089\pi}$. Find α . Speed of sound $v = 330 \text{ m/s}$, density of air $\rho_0 = 1.0 \text{ kg/m}^3$.
6. The current through 12Ω resistance is zero. The value of E (in Volt) is :



Space for Rough Work

SECTION-III : (Maximum Marks : 18)

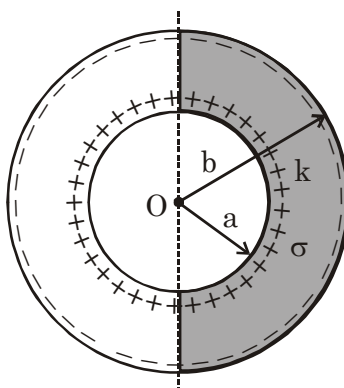
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- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : -1 In all other cases

1. A particle having specific charge σ is projected in xy plan with a speed v . There exists a uniform magnetic field in z-direction having a fixed magnitude B_0 . The field is made to reverse its direction after every interval of $\frac{2\pi}{\sigma B_0}$. Calculate the maximum separation (in m) between two positions of the particle during its course of motion. (Given $\frac{v}{\sigma B_0} = 2$ meter) (neglect any other force including gravity throughout the motion)
2. Half portion of a spherical capacitor is filled with a dielectric of dielectric constant $k = 2$ conductivity σ . The charge given to spherical capacitor is Q_0 . Due to the conductivity of dielectric charge leaks and the time constant for the discharge circuit is x time of $\frac{\epsilon_0}{\sigma}$. Find the value of x .

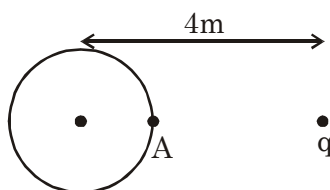


Space for Rough Work

3. A parallel plate capacitor (square plates) has a capacitance of 1 mF without the dielectric. It is completely filled with a solid dielectric (constant = 5) which is performing very small oscillations (SHM, of amplitude 0.1 % of plate dimension) with a period of 1 ms as shown : At all times, a battery of 1 V remains connected to the capacitor. If the amount of charge that crosses the battery. When the dielectric goes from extreme to mean is Δq μC , find value of Δq . Assume that oscillation is caused by an external agent.



4. A particle of mass 1 kg and charge 0.5 C, is projected in a magnetic field of magnitude 2T, with a velocity 3 m/s making an angle of 30° with the field. Find radius of curvature of path traced out by the particle (in metres) [neglect gravity].
5. Electric field due to a uniformly charged ring of radius $\sqrt{5}$ m at a certain point on its axis is 1 V/m and electric potential at the same point is 6V. If the distance of this point from centre of the ring is x (in m), find the smallest value of x.
6. A neutral conducting solid sphere of radius 2m is placed such that its centre is 4m from a point charge $q = 2\text{mC}$. The electric field intensity just outside at a point A on the sphere is found to be 2 N/C. If the induced charges per unit area at A is $x\epsilon_0$. Find the value of x.

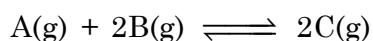


Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I : (Maximum Marks: 24)**

- This section contains **SIX** questions.
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- For each question, choose the correct option(s) to answer the question.
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. Reactants and products of a reaction are at equilibrium as :



In a closed rigid container if gas 'A' is added to the equilibrium mixture then -

- (A) Concentration of 'A' at new equilibrium is greater than old equilibrium
 - (B) Concentration of 'B' at new equilibrium is less than old equilibrium
 - (C) Concentration of 'C' at new equilibrium is more than old equilibrium
 - (D) Value of K_c and K_p are more at new equilibrium than old equilibrium
2. Select the **CORRECT** statement(s) -
- (A) On increasing temperature, conductivity of semiconductors increases.
 - (B) Non-stoichiometric Cu_2O can act as p-type semiconductor
 - (C) MnO is paramagnetic solid at room temperature.
 - (D) HCl is molecular solid

Space for Rough Work

3. Select the **CORRECT** statement(s)?

- (A) P–O bond length in POF_3 is more than P–O bond length in PO_3F^{2-}
 (B) P–O bond length in PO_4^{3-} is more than P–O bond length in PO_3F^{2-}
 (C) O–P–O bond angle in PO_4^{3-} is less than O–P–O bond angle in PO_3F^{2-}
 (D) P–F bond length in POF_3 is more than P–F bond length in PO_3F^{2-}

4. _____ has a/an _____ structure.

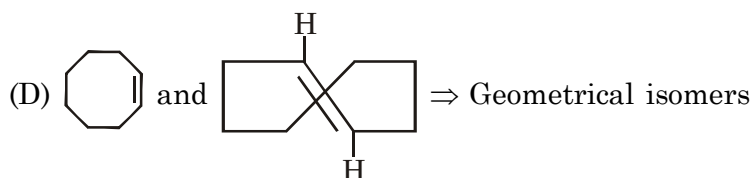
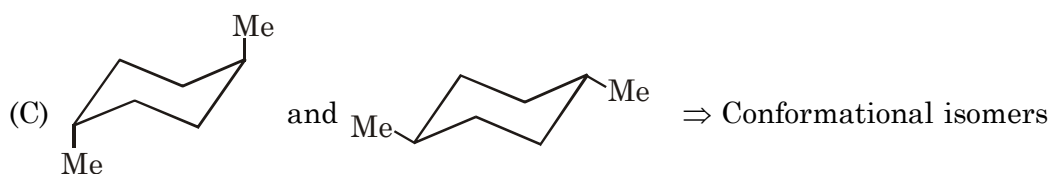
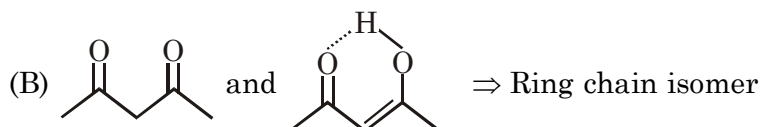
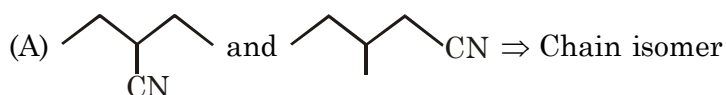
Choose the **CORRECT** option(s) which can fill correctly in the blanks.

- (A) Rhombic sulphur ; crown (B) Black phosphorous ; discrete molecular
 (C) Hydrogen peroxide ; planar (D) IO_2F_2^- , see-saw

5. Select reaction with $K_{eq} < 1$

- (A) $\text{PhOH} + \text{NaHCO}_3 \rightleftharpoons \text{PhONa} + \text{H}_2\text{O} + \text{CO}_2$
 (B) $\text{Me} - \text{C} \equiv \text{CH} + \text{NaOMe} \rightleftharpoons \text{MeC} \equiv \text{CNa}^\oplus + \text{MeOH}$
 (C) $\text{EtOH} + \text{NaNH}_2 \rightleftharpoons \text{EtONa} + \text{NH}_3$
 (D) $\text{H}_2\text{O} + \text{NaCl} \rightleftharpoons \text{HCl} + \text{NaOH}$

6. Correctly matched pair out of the following is/are :



Space for Rough Work

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+									
●	●	○	○	○	○	○	○	○	○
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	○	○	○	○	○	○	○
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

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- 32 gm of a hydrated salt $\text{ASO}_4 \cdot x\text{H}_2\text{O}$, when dissolved in 84 gm of water, the solution freezes at -4.0°C . If $K_f = 1.8 \text{ K}\cdot\text{kg}/\text{mole}$ for water and ASO_4 is a strong electrolyte. Determine the number of mole of H_2O per mole of salt. [Given molar mass of salt ASO_4 is 120]
- If 10^{23} radioactive atoms are added after each half life in a sample of 10^{24} radioactive atoms, then number of atoms disintegrated between 2nd and 3rd half life is $x \times 10^{23}$
- Consider BrF_5 molecule and calculate $x + y + z$
 $x \rightarrow$ number of maximum atoms present in a plane
 $y \rightarrow$ number of d-orbitals involved in hybridisation
 $z \rightarrow$ number of total lone pairs

Space for Rough Work

4. $\underline{\text{B}}_2\text{H}_6$, $\underline{\text{Si}}_2\text{H}_6$, $\underline{\text{Al}}_2\text{Cl}_6$, $\underline{\text{Be}}_2\text{H}_4$, $\underline{\text{I}}_2\text{Cl}_6$, $\underline{\text{B}}\text{H}_3\cdot\text{NH}_3$, $\cdot\underline{\text{C}}\text{F}_3$, $\cdot\underline{\text{C}}\text{H}_3$

Condition (1) : Species in which electron geometry at underlined atom is tetrahedral

Condition (2) : Species in which 3c-2e⁻ bond are present

Condition (3) : Atleast four atoms are present in a plane

then find multiplication of x and y

Here x = number of species which is/are satisfying condition (1) and (3) both

y = number of species which is/are satisfying either condition (2) or (3)

5. Calculate the number of Benzenoid isomers possible for $\text{C}_6\text{H}_3\text{ClBrI}$.
6. A mixture of d and *l*, 2-bromobutane contain 75% d-2-bromobutane. Calculate % enantiomeric excess.

Space for Rough Work

SECTION-III : (Maximum Marks : 18)

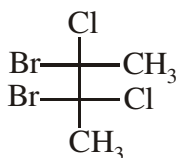
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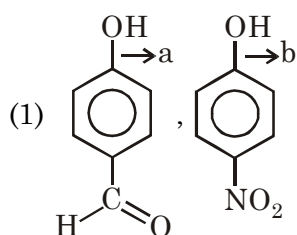
Negative Marks : -1 In all other cases

1. 0.8 gram-atoms of an α -emitting radionuclide A_ZX ($t_{1/2} = 2$ hours.) producing ${}^{A-4}_{Z-2}Y$ were placed in a sealed container. If 15.68 litres of He(g) was collected in container kept in 1atm 273K conditions in t hours. Find value of ' t ' ?
2. The activation energy of a second order reaction : $A + B \rightarrow P$ is $9.2 \text{ kcal mole}^{-1}$. At 500K product is formed at the rate of $0.08 \text{ mol L}^{-1}\text{min}^{-1}$. Find the rate of formation of product at 1000 K in $\text{mol L}^{-1}\text{min}^{-1}$. if other factors are assumed to be uniform [Use $\ln x = 2.3 \log x$]
3. Find out the number of pairs in which boiling point of first compound is less than second? $(B(Me)_3, BF_3)$; $(SiCl_4, CCl_4)$; (H_2, O_2) ; $(NF_3, N(Me)_3)$; $(H_3BO_3, B(OMe)_3)$; (CO, N_2) ; (H_2O, HF)
4. Find the number of compounds from the following in which 6 membered ring is/are present. S_3O_9 , White phosphorus, P_4O_7 , $C_3N_3Cl_3$, Graphite, Borax, Benitotite $[BaTi(Si_3O_9)]$, C_{60} (Fullerene)
5. The total number(s) of **stable** conformers with **non-zero** dipole moment for the following compound is (are)

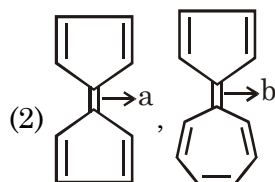


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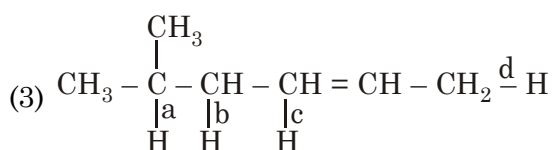
6. How many properties given are correct :



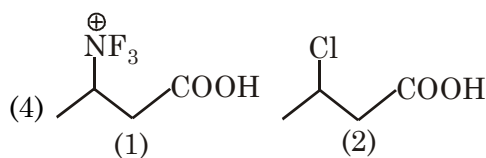
Bond length : $a > b$



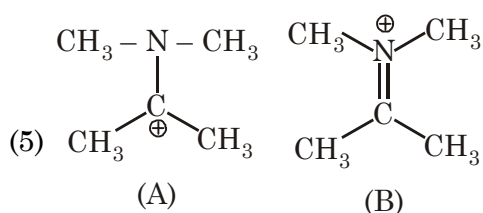
Rotational energy barrier : $a > b$



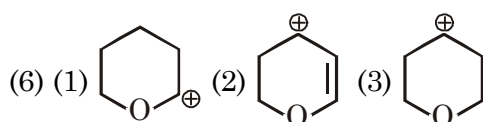
C-H bond energy : $c > b > a > d$



Acidic strength : $2 > 1$



Stability : $B > A$



Stability $1 > 2 > 3$

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PART-3 : MATHEMATICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. If $L = \lim_{x \rightarrow 0} \frac{a \sin x - bx + cx^2 + x^3}{2x^2 \log(1+x) - 2x^3 + x^4}$, where L is a finite quantity, then -
- (A) $a = 6$ (B) $b = -6$ (C) $L = \frac{3}{40}$ (D) $c = 0$
2. A function is defined as $f(x) = [\sec x] + \sqrt{\sec x - [\sec x]}$, $0 \leq x < \frac{\pi}{2}$ (where $[.]$ denotes greatest integer function), then
- (A) $f(x)$ is continuous at $x = 0$ (B) $f(x)$ is discontinuous at $x = \frac{\pi}{3}$
- (C) $f(x)$ has infinite points of discontinuities (D) $f(x)$ is continuous for all $x \in \left[0, \frac{\pi}{2}\right)$
-

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3. If $f(x) = \max \{1 + \sin x, 3 - \cos x, 2\}$ & $g(x) = \min \{x^2, 1 - x, 4x - 3\}$, then -
 (A) $\text{fog}(0) = 2$ (B) $\text{fog}(0) > 2$ (C) $\text{gof}(0) = -1$ (D) $\text{gof}(0) < -1$
4. Let $f : [-1, 1] \rightarrow [-1, 1]$, $f(x) = \sin(2\sin^{-1}x)$ & let $g(x) = \frac{f(x)}{\sqrt{1-x^2}}$, then -
 (A) $f(x)$ is one-one function (B) $f(x)$ is odd function
 (C) $g(x)$ is continuous in $(-1, 1)$ (D) $f(x)$ is continuous for all $x \in [-1, 1]$
5. If three lines, $x + 2y - 4 = 0$, $3x - ky + 2 = 0$ and $5x + 3y + 1 = 0$, do not form a triangle, then k may be equal to
 (A) $-\frac{4}{3}$ (B) -6 (C) $-\frac{9}{5}$ (D) $-\frac{6}{5}$
6. If $f(x) = 2x^2 + kx + 50$, $k \geq 20$ and magnitude of difference of the roots of the equation $f(x) = 0$ is 8, then which of the following can be true -
 (A) minimum value of $f(x)$ is -32 (B) minimum value of $f(x)$ is -64
 (C) graph of $y = f(x)$ cuts positive y-axis (D) graph of $y = f(x)$ cuts negative y-axis

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	●	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Let $f(x)$ be a polynomial of degree 2005 with leading coefficient 2006 such that for $n = 1, 2, 3, \dots, 2005$; $f(n) = n$. Then the number of zeros at the end of $f(2006) - 2006$ is
- In a right angled triangle ABC, right angled at A, the radius of inscribed circle is 2 and radius of escribed circle touching side BC is 15, then length of side BC is
- The value of $\sin 5^\circ \cos 15^\circ \sin 55^\circ \sin 65^\circ$ is

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4. If $|x| + y = 3$ and $\left| \sin \frac{\pi x^2}{2} \right| = 1$ then number of ordered pairs (x, y) is (where y is a non negative integer)

5. Consider the curve $f(x) = x \sin x$. At variable point of curve whose abscissa is x_1 , tangent of slope $(\sin x_1 + x_1 \cos x_1)$ is drawn. If the sum of intercepts of tangent on coordinate axes is $\mu(x_1)$,

then $\left| \lim_{x_1 \rightarrow 0} \left(\frac{\mu(x_1)}{x_1} \right) \right|$ is

6. Let $f(x) = \begin{cases} \ln((\cos x + a \sin bx)^{1/x}); & x \in \left(0, \frac{\pi}{2}\right) \\ 2 & ; \quad x = 0 \\ b - 2x & ; \quad -1 < x < 0 \\ 5 - cx & ; \quad x \leq -1 \end{cases}$ and $f(x)$ is continuous in its domain.

If $\cos^{-1} \cos(a + c + 3b) = m\pi + n$, then value of $(m - n)$ is-

Space for Rough Work

SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. The value of $\lim_{x \rightarrow 0} \frac{(\sin(\ell n e^x))^2}{(e^{\tan^2 x} - 1)}$ is

2. If $f(x) = \begin{cases} \frac{\sin(a+2)x + \sin x}{x} ; & x < 0 \\ b & ; \quad x = 0 \\ \frac{(x+3x^2)^{1/3} - x^{1/3}}{x^{4/3}} ; & x > 0 \end{cases}$ is continuous at $x = 0$ then $|a + b|$ is

3. If $\sum_{r=1}^{\infty} \cot^{-1} \left(\frac{(r+1)^2}{2} \right) = \tan^{-1} x$, then x is equal to -

Space for Rough Work

4. Number of elements in the domain set of the function $f(x) = \tan^{-1} \sqrt{[x] + [-x]} + \sqrt{2-|x|} + \frac{1}{x^2}$ is (where $[*]$ is the greatest integer function)
5. The value of 'p' for which the given system $x + py + 2z = 5$, $2x + y + 2z = 4$ and $x + y + 2z = 3$ is inconsistent, is
6. If positive numbers a, b, c, d are in A.P. such that $(ab + bc + cd)^2 \geq (a^2 + b^2 + c^2)(b^2 + c^2 + d^2)$, then
$$\frac{\log_e (c+d)^{(a+b)} + \log_e (d+a)^{(b+c)} + \log_e (a+b)^{(c+d)} + \log_e (b+c)^{(d+a)}}{\log_e (a+b)^a + \log_e (b+c)^b + \log_e (c+d)^c + \log_e (d+a)^d}$$
 is equal to (wherever defined)

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has three section as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
II	Numerical Value Type (Up to second decimal place)	6	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24
III	Single digit Integer (0-9)	6	+3 If only the bubble corresponding to correct answer is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	18

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator